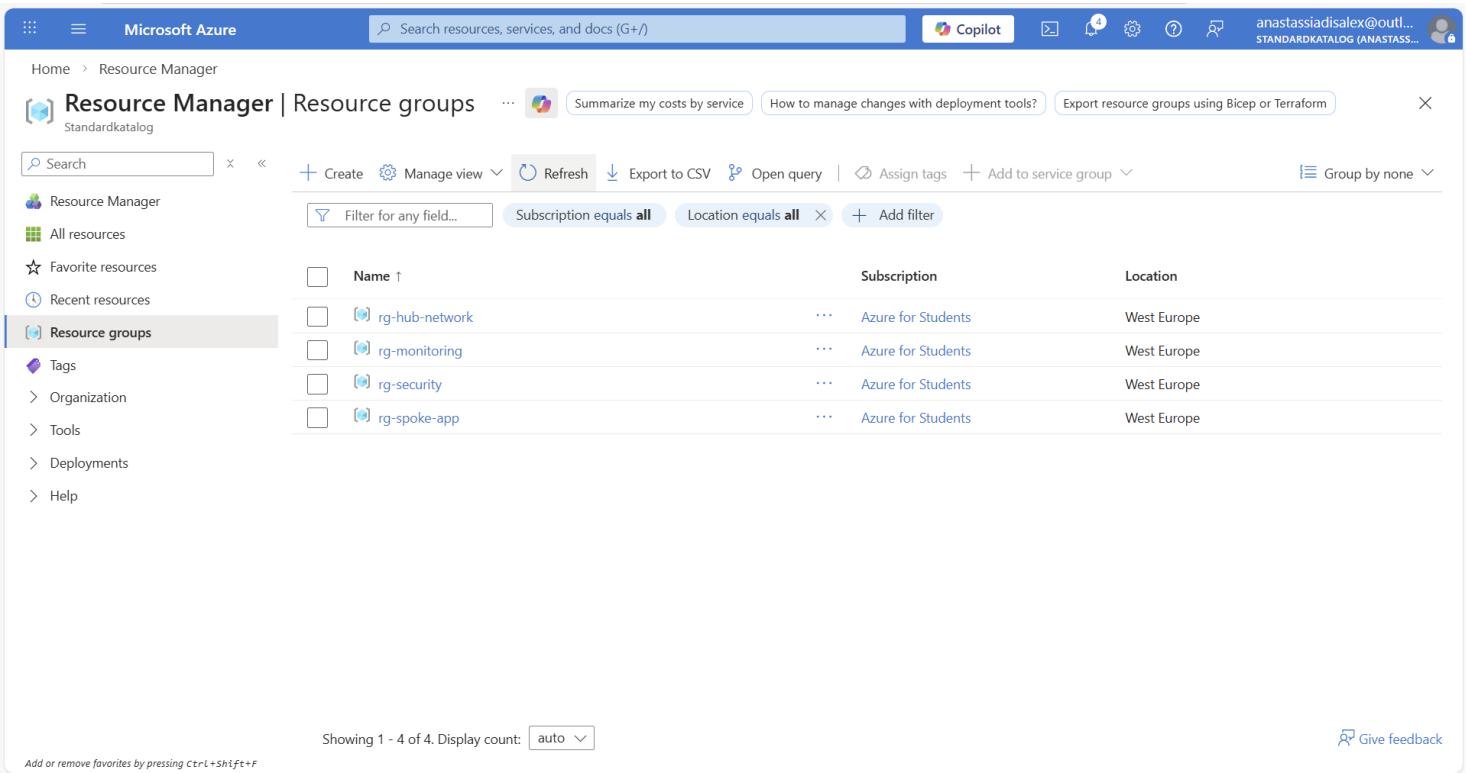
# Azure Enterprise Secure Lab

## Phase 1 – Secure Networking Foundation

*Step 1.1 – Create Resource Groups*

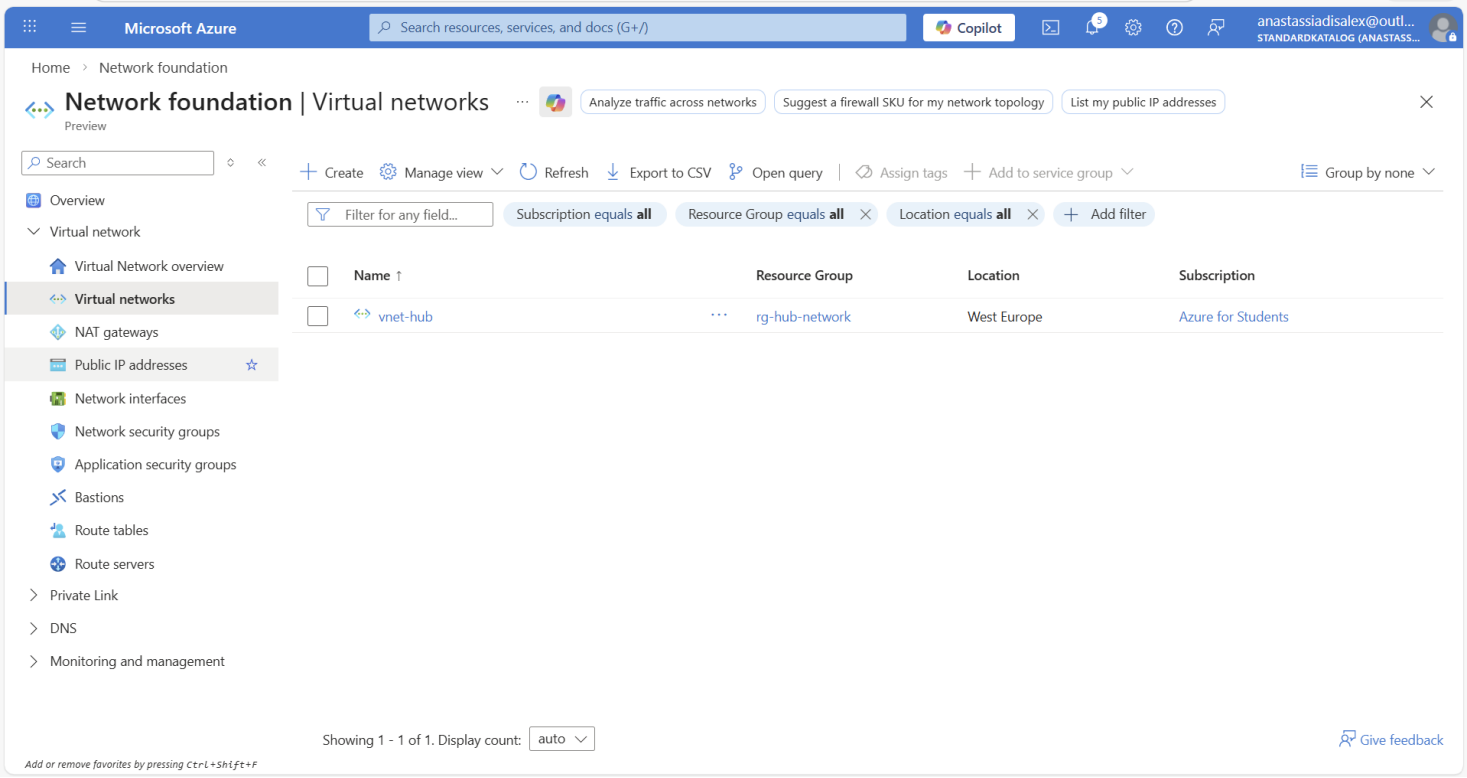
Creating four different resource groups to seperate and make the infrastructure as clean as possible, this makes RBAC Role-Based Accessed Control and to isolate the resources.

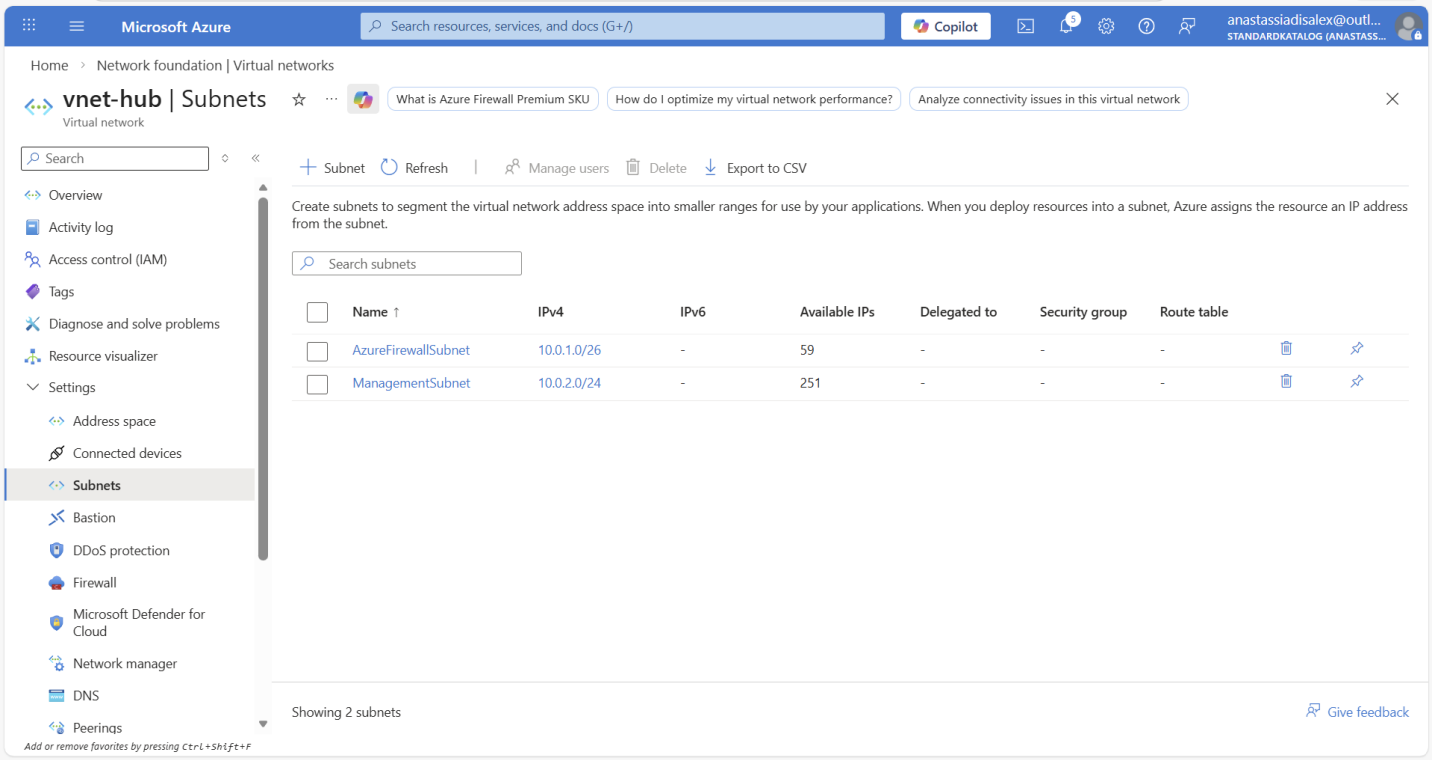



*Step 1.2 – Create Hub VNet*

A Hub Virtual Network was made to host different devices, and two subnets were made.

- (AzureFirewallSubnet & ManagementSubnet)

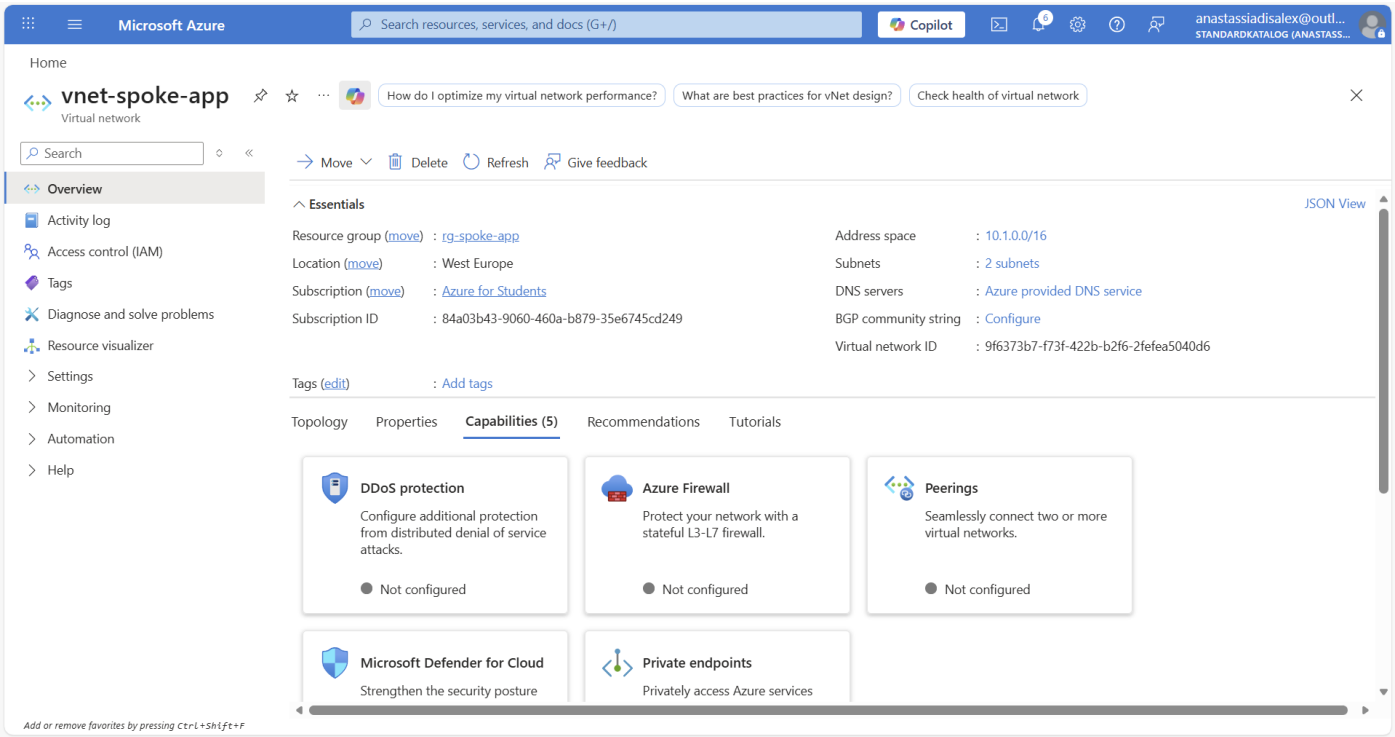


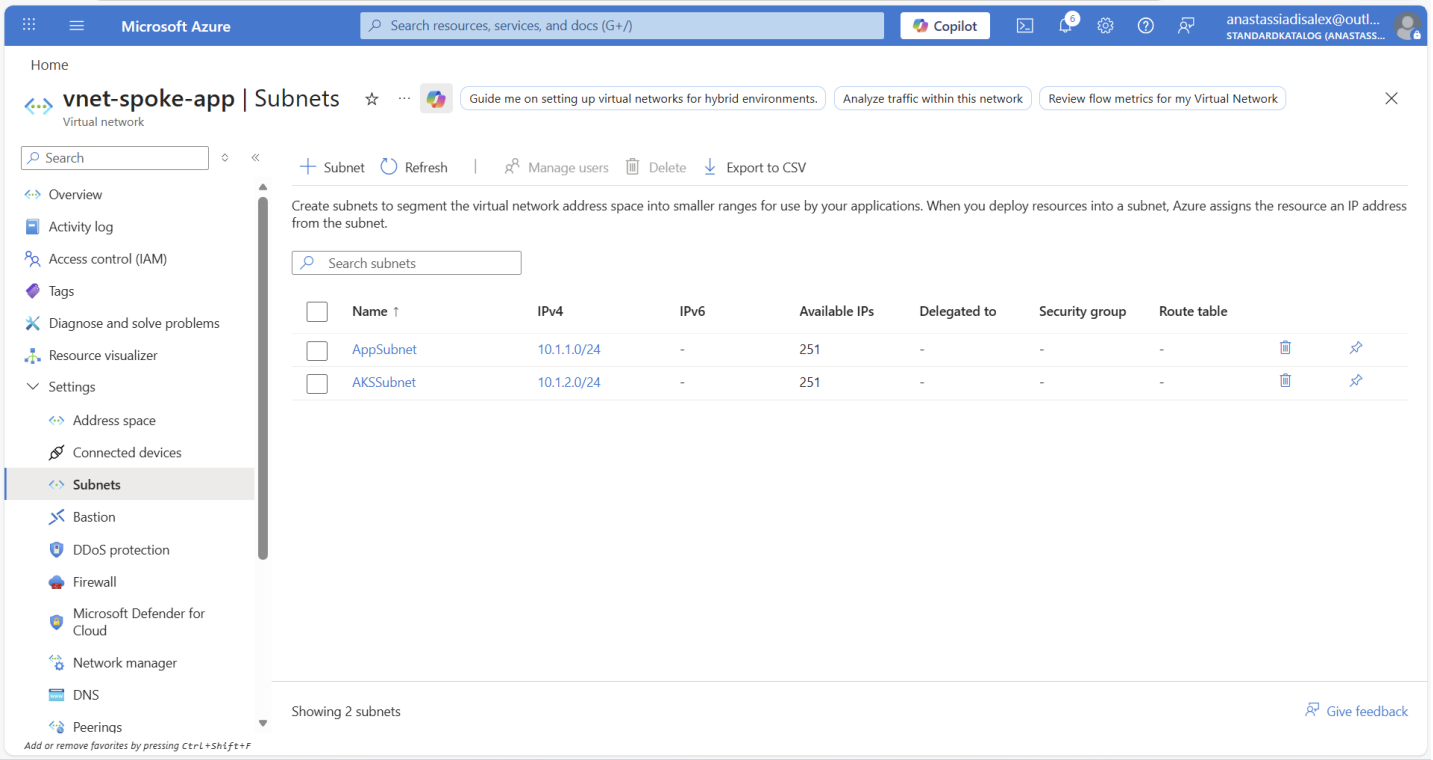




*Step 1.3 – Create Spoke VNet*

Another Virtual Network was made (vnet-spoke-app) to host application workload

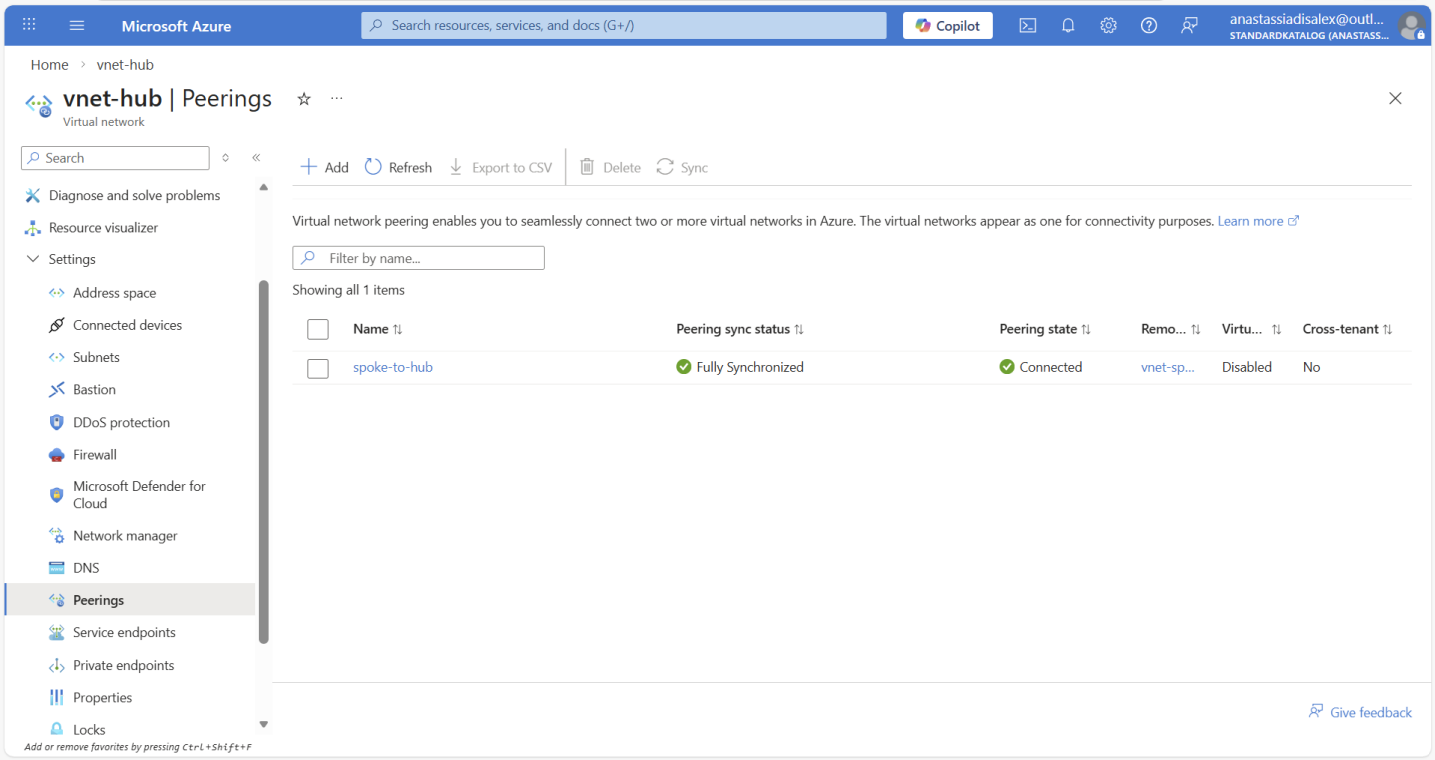
Also created two subnets within (AppSubnet & AKSSubnet)

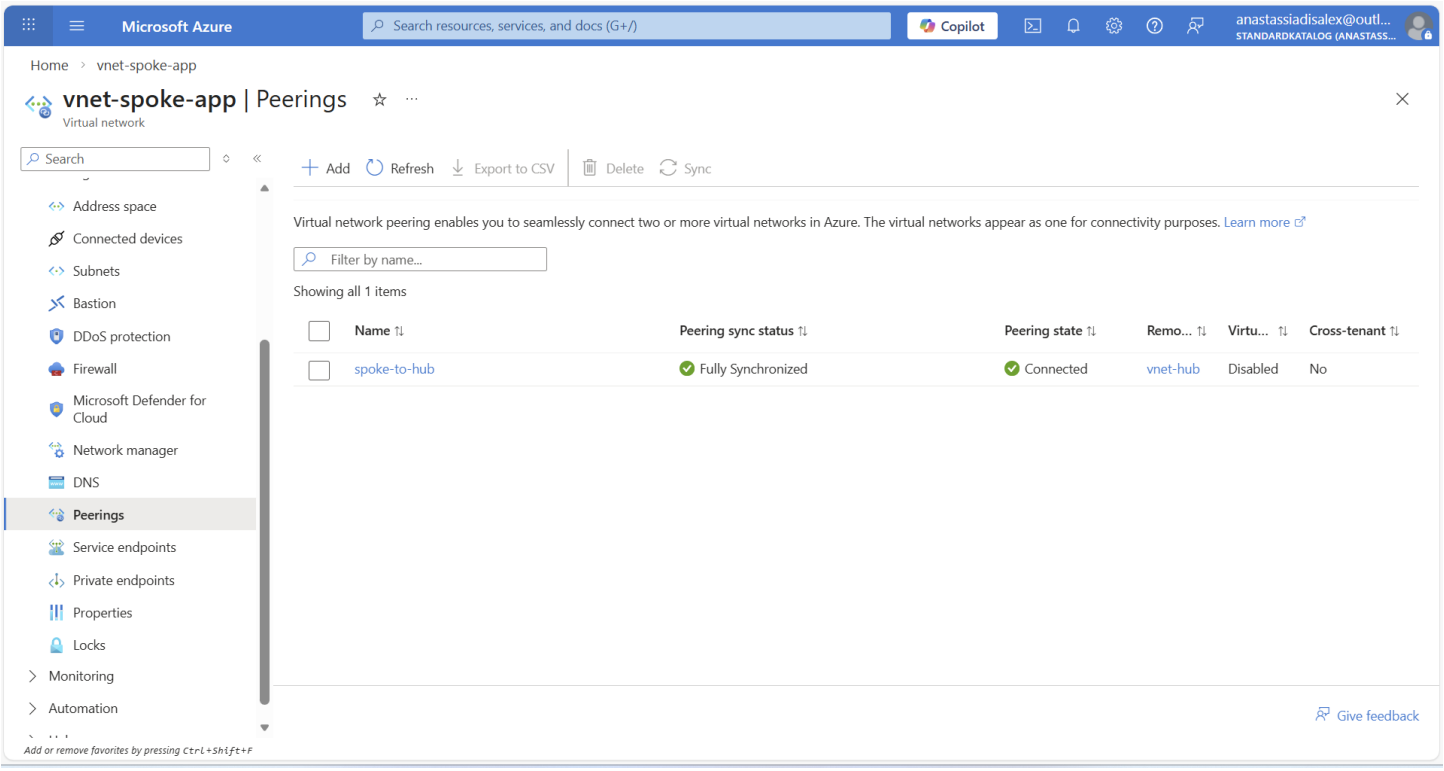



*Step 1.4 – VNet Peering*

Peering VNet Hub --> VNet Spoke and the other way around. Azure creates the peering automatically with the same name as its peer one-way that's why it’s the same name on the peering. But it's shown in the images below that it's connected to both VNets. It’s making a secure communication between the two networks.



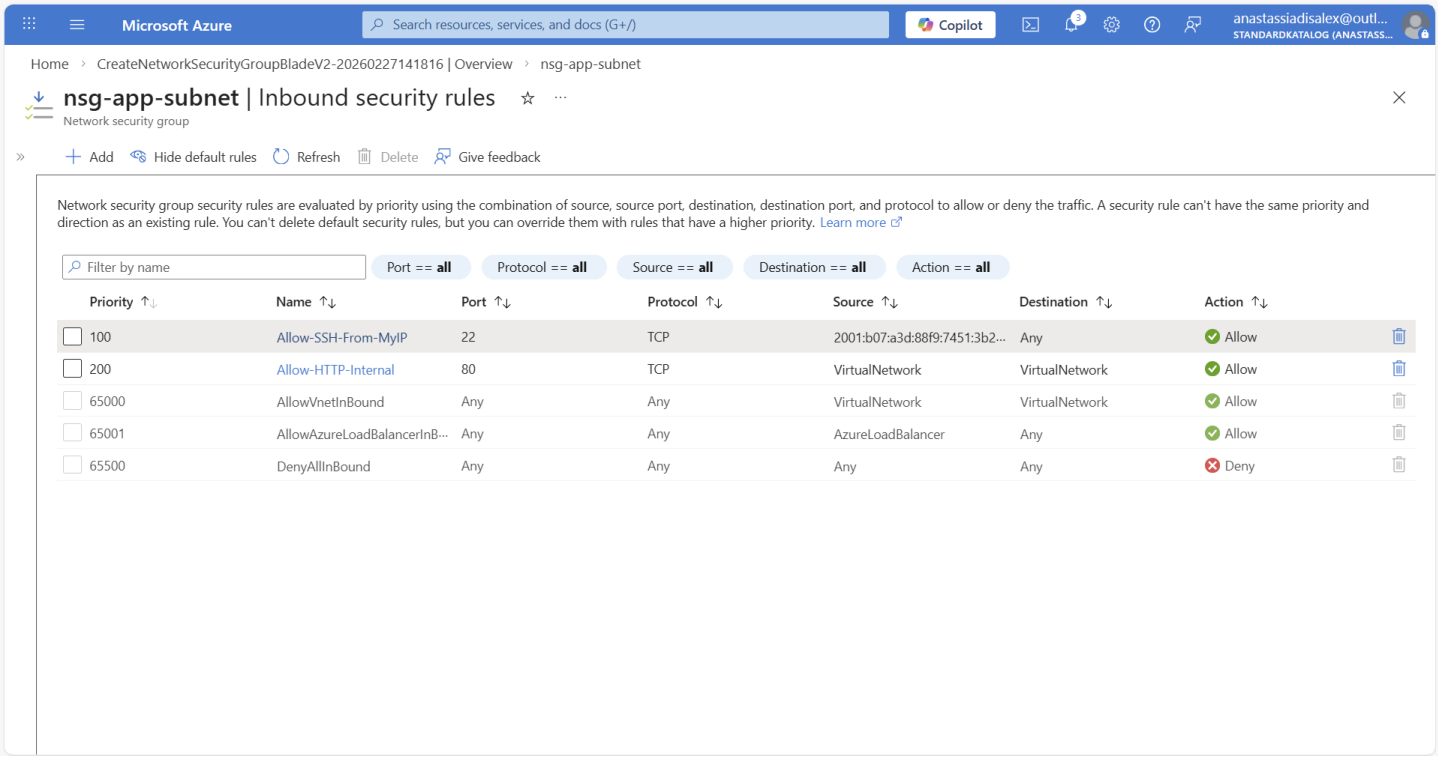



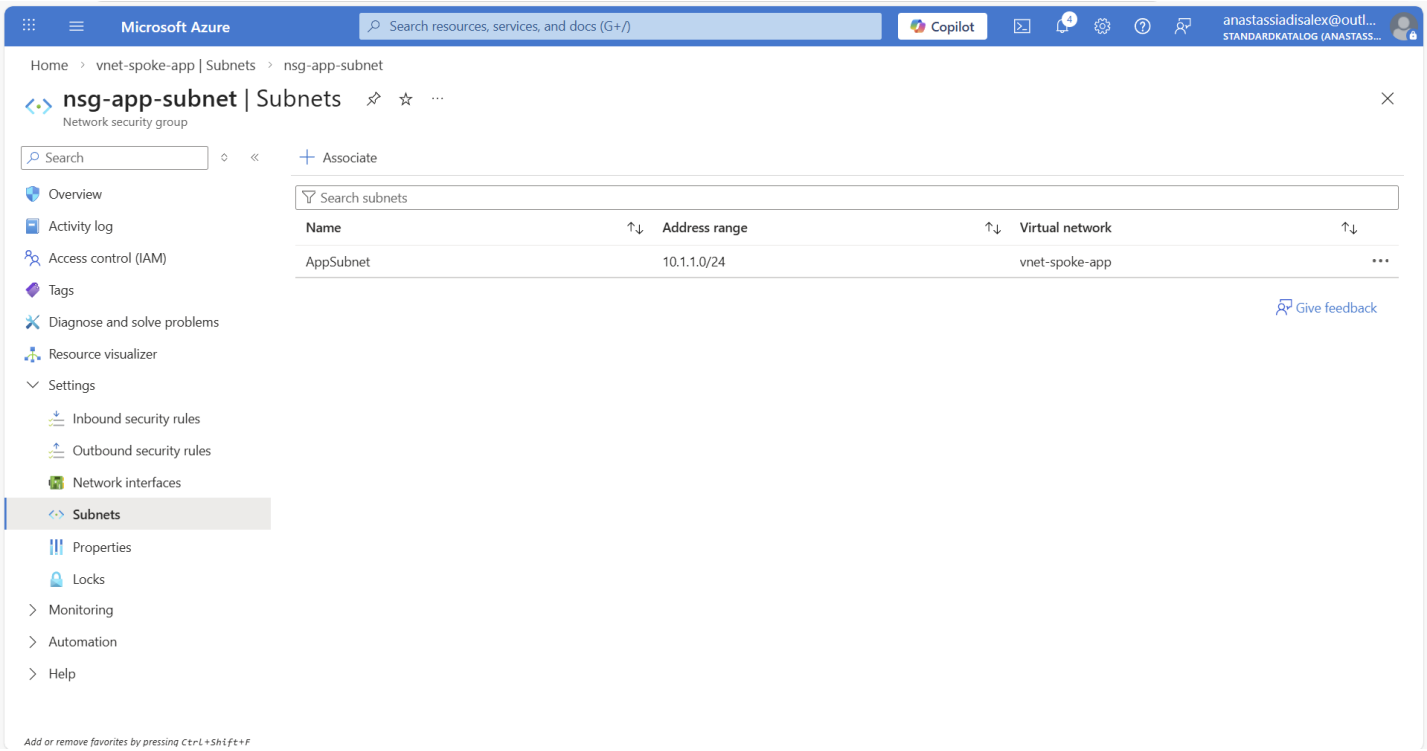

*Step 1.5 – Network Security Groups*

A NSG created ans associated to the subnet (AppSubnet)

Created rules (Allow SSH from my IP & Allow HTTP Internally) As well as Deny All Inbound which is default.

This enforces least privilege network access and prevents unauthorized exposure.

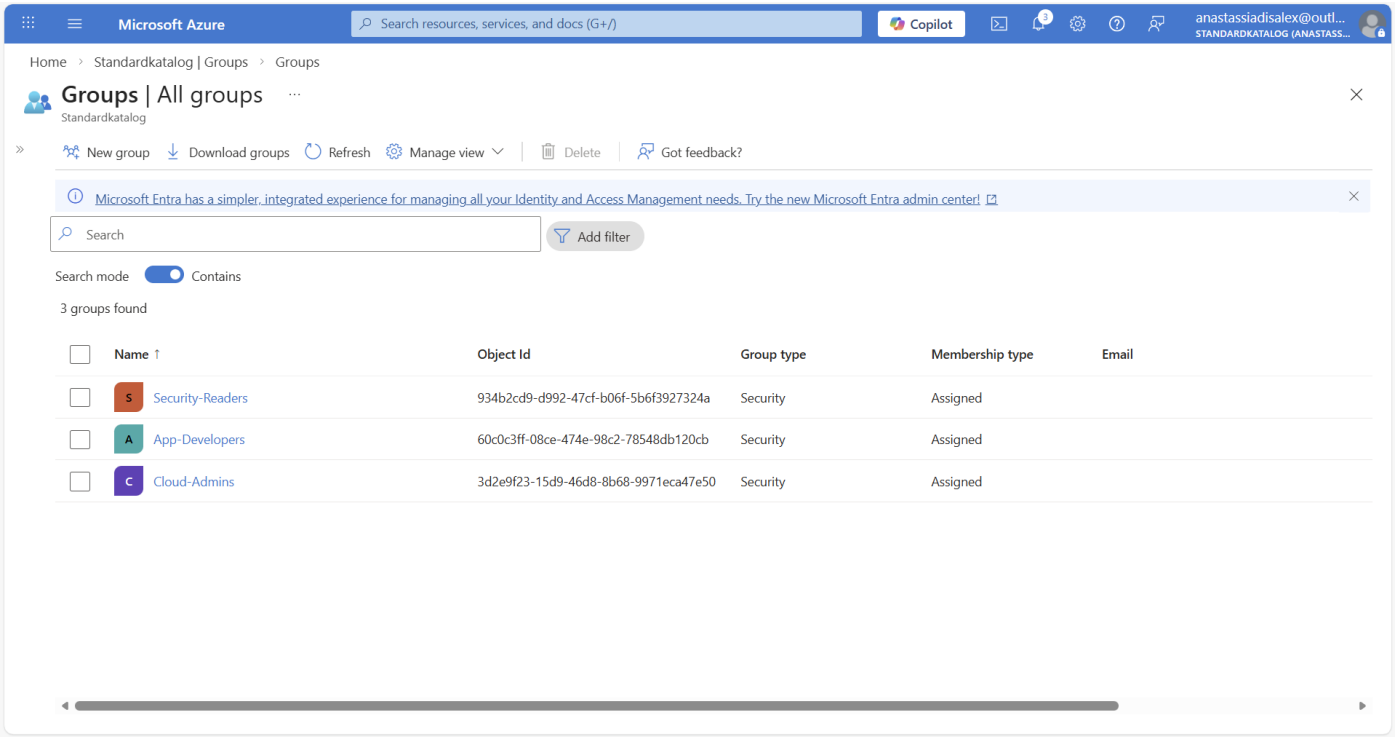



## Phase 2 – Identity & Governance

*Step 2.1 – Entra ID (RBAC Design)*

Created groups in Entra ID and assigned appropriate roles to one another. This enforces role-based access control following the principle of least privilege.




Giving roles in IAM.

Reader + Security Reader to Security-Readers on subscription level.

Contributor to App-Developers (in RG rg-spoke-app).

Owner role to Cloud-Admins on a Subscription level.

Microsoft Azure | Search resources, services, and docs (G+/) | Copilot | anastassiadisalex@outl... STANDARDKATALOG (ANASTASS...

Home > Azure for Students

# Azure for Students | Access control (IAM)

Subscription

Search

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Add | Download role assignments | Edit columns | Refresh | Delete | Feedback

Co-Administrator or Service Administrator role assignments, convert these roles to Azure RBAC immediately. Learn more

Check access | Role assignments | Roles | Deny assignments | Classic administrators

**Number of role assignments for this subscription**

12 4000

**Privileged**

3

View assignments

Security-Readers | Type : **All** | Role : **All** | Scope : **All scopes** | Group by : **Role**

All (2) | Job function roles (2) | Privileged administrator roles (0)

| Name | Type | Role | Scope | Condition |
|---|---|---|---|---|
| **Reader (1)** | | | | |
| SR Security-Readers 934b2cd9-d992-47cf-b0... | Group | Reader | This resource | None |
| **Security Reader (1)** | | | | |
| SR Security-Readers 934b2cd9-d992-47cf-b0... | Group | Security Reader | This resource | None |

Showing 1 - 2 of 2 results.

*Add or remove favorites by pressing Ctrl+Shift+F*

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Home > Resource Manager | Resource groups > rg-spoke-app

# rg-spoke-app | Access control (IAM)

Resource group

Search

- Overview
- Activity log
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- Tags
- Resource visualizer
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- Cost Management
- Monitoring
- Automation
- Help

Add | Download role assignments | Edit columns | Refresh | Delete | Feedback

Check access | Role assignments | Roles | Deny assignments | Classic administrators

**Number of role assignments for this subscription**

10 4000

**Privileged**

4

View assignments

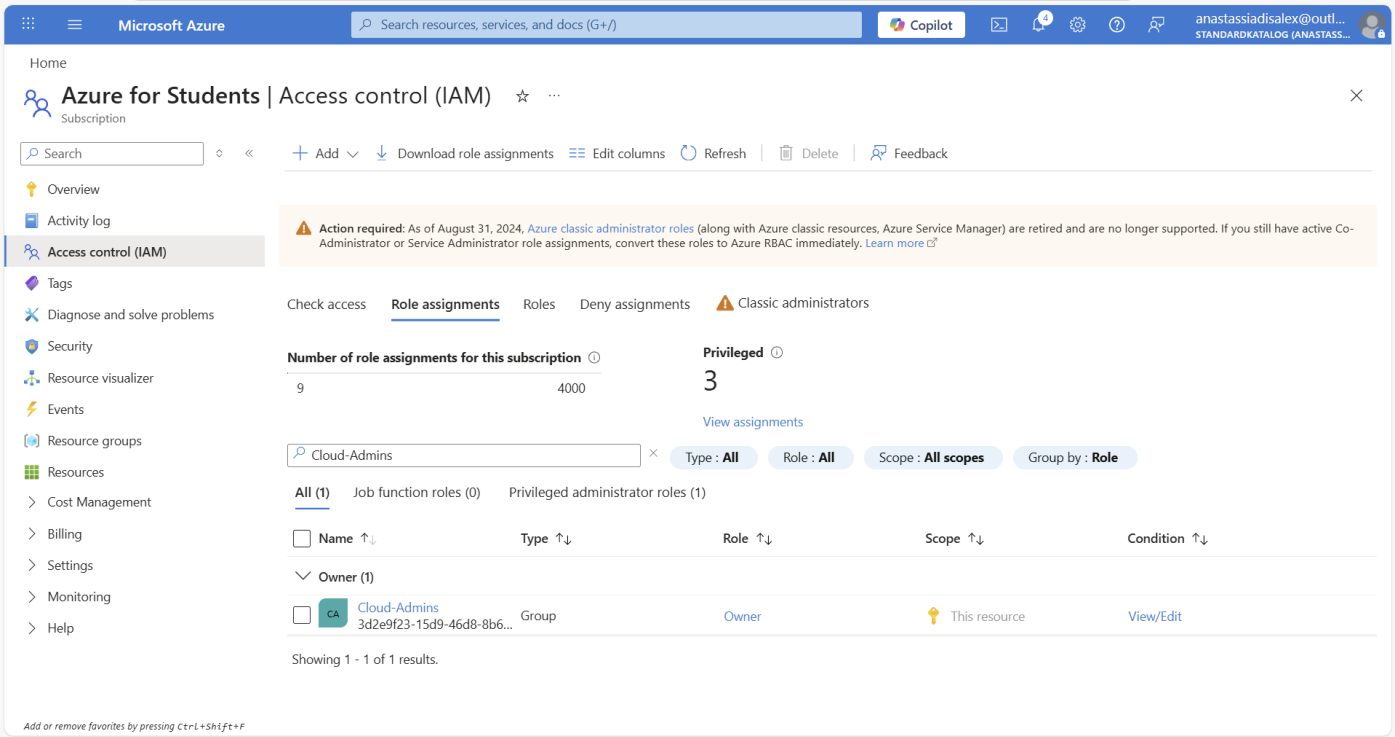
App-Developers | Type : **All** | Role : **All** | Scope : **All scopes** | Group by : **Role**

All (1) | Job function roles (0) | Privileged administrator roles (1)

| Name | Type | Role | Scope | Condition |
|---|---|---|---|---|
| **Contributor (1)** | | | | |
| AD App-Developers 60c0c3ff-08ce-474e-98c2... | Group | Contributor | This resource | None |

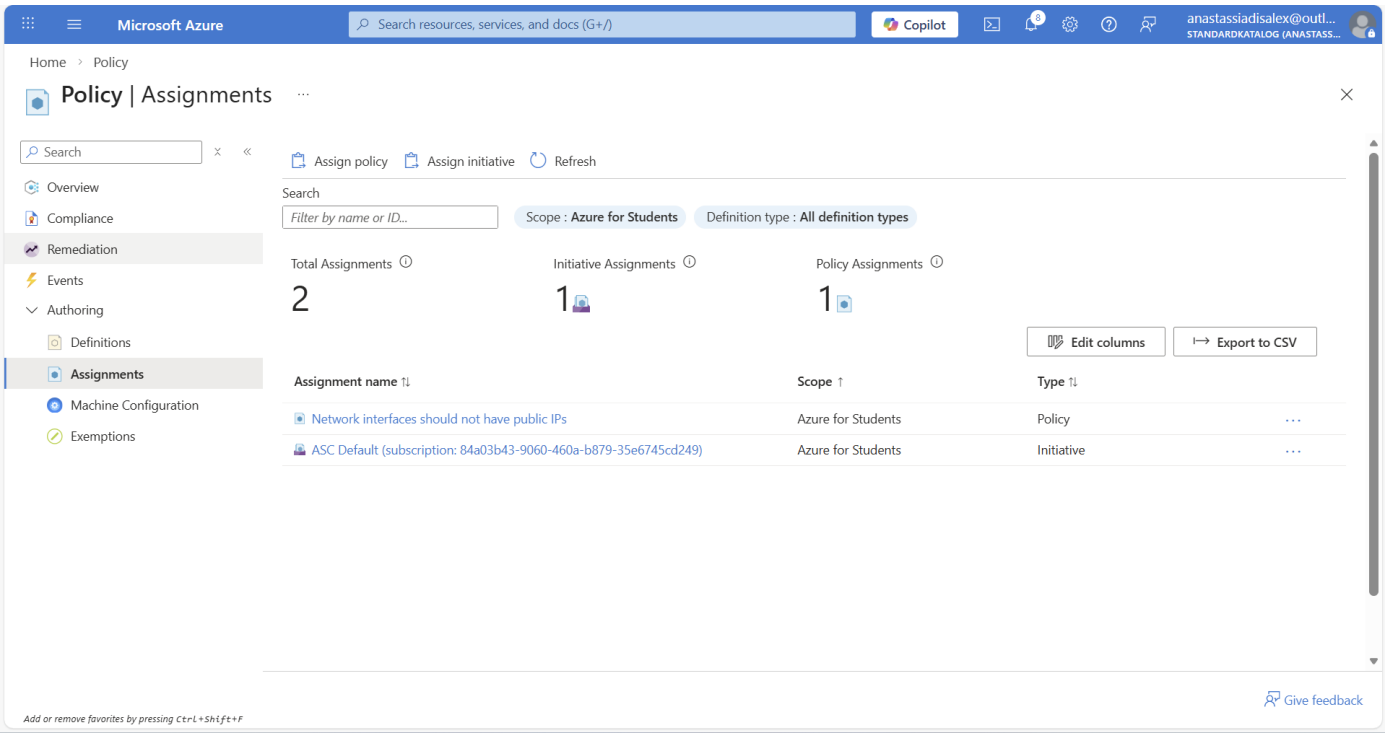
Showing 1 - 1 of 1 results.

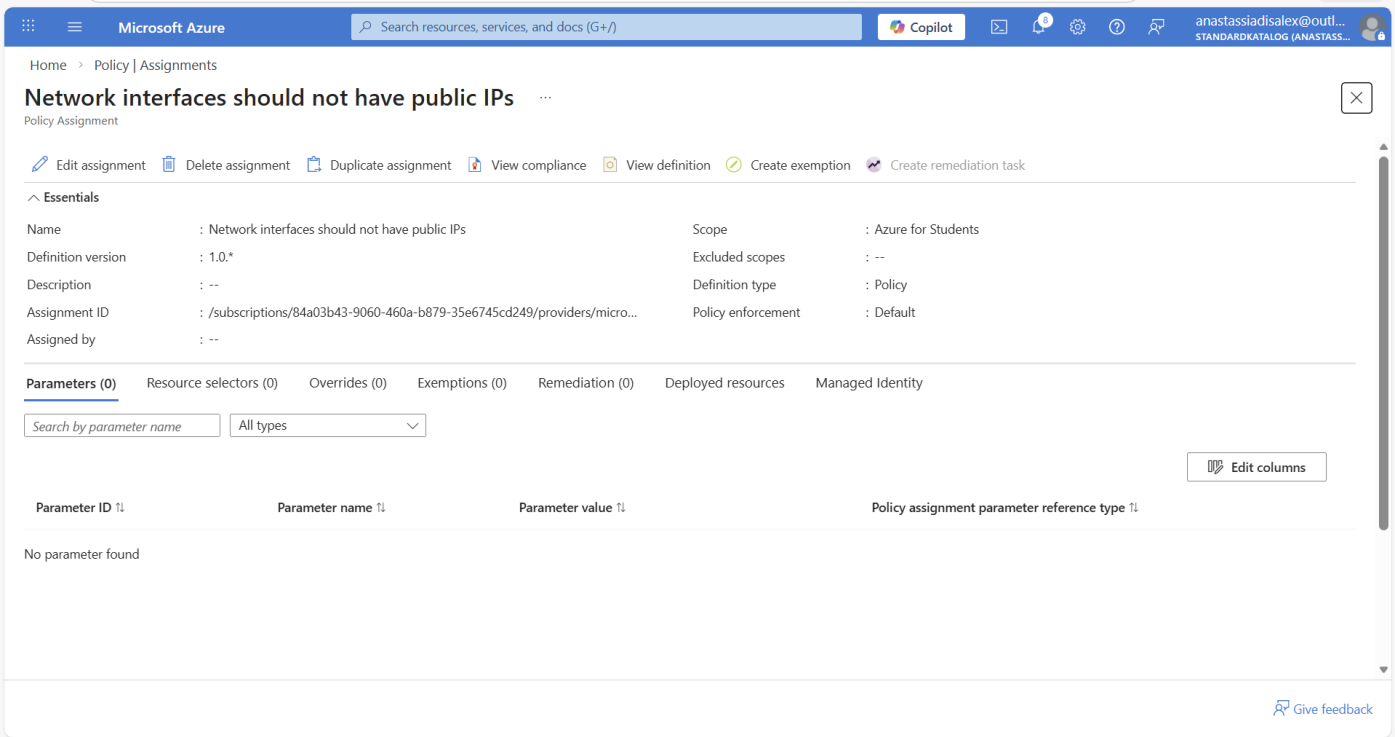
*Add or remove favorites by pressing Ctrl+Shift+F*



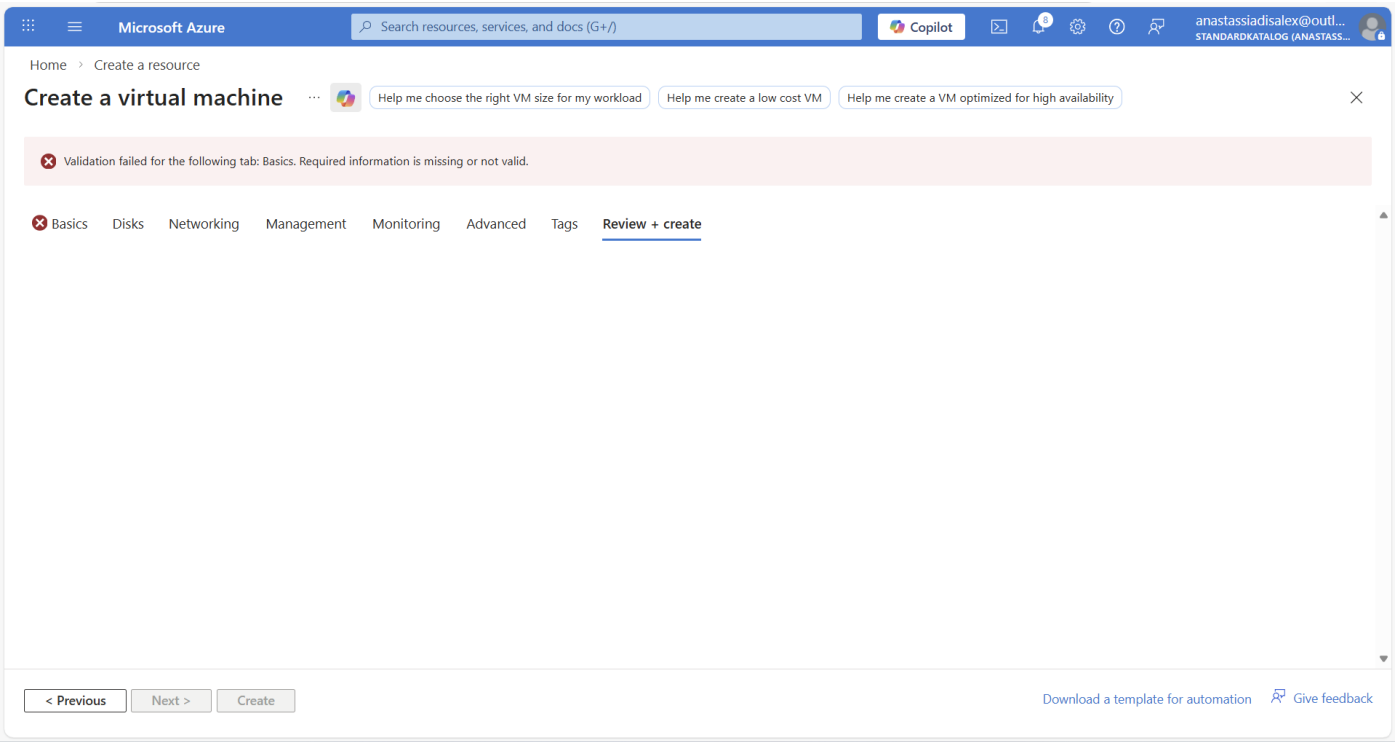

## *Step 2.2 – Azure Policy*

Policy assignment (Network interfaces should not have public IPs)

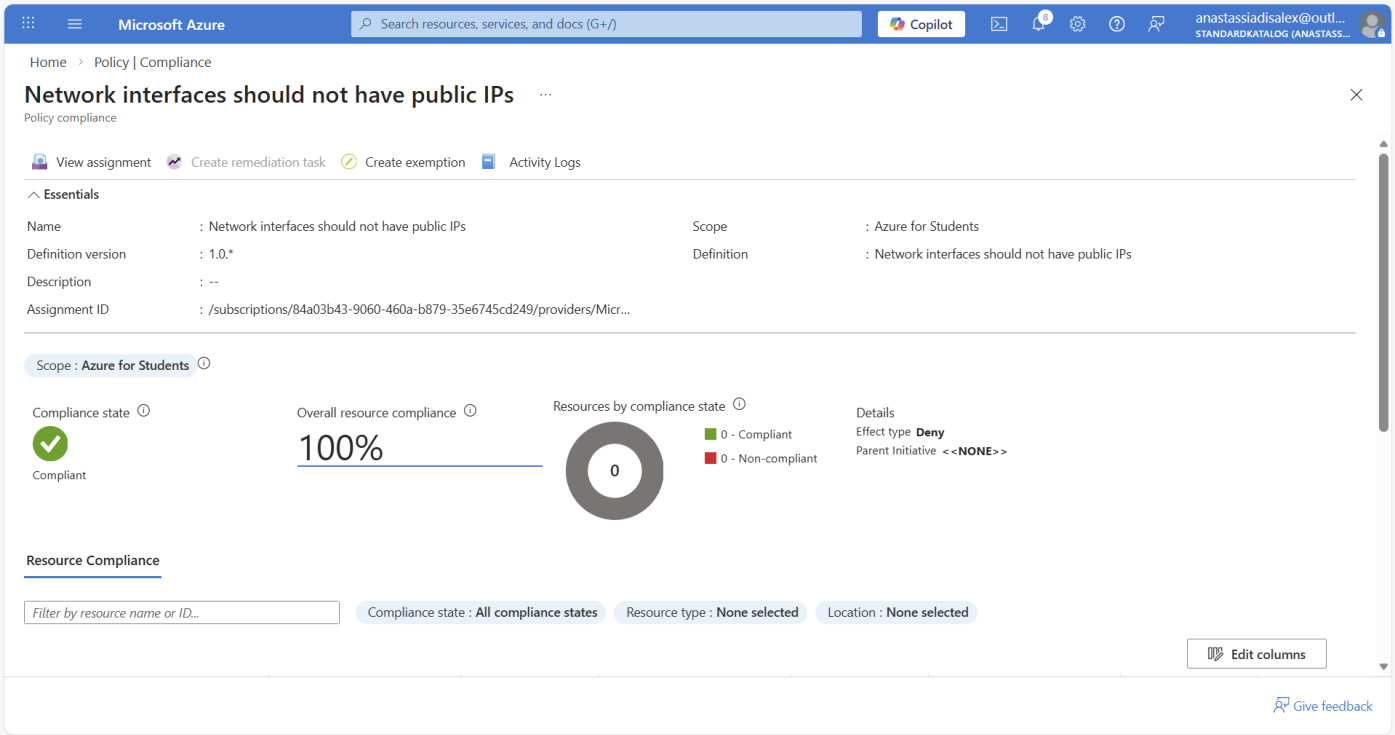





Testing to create a VM with Public IP (shows error because of the policy)



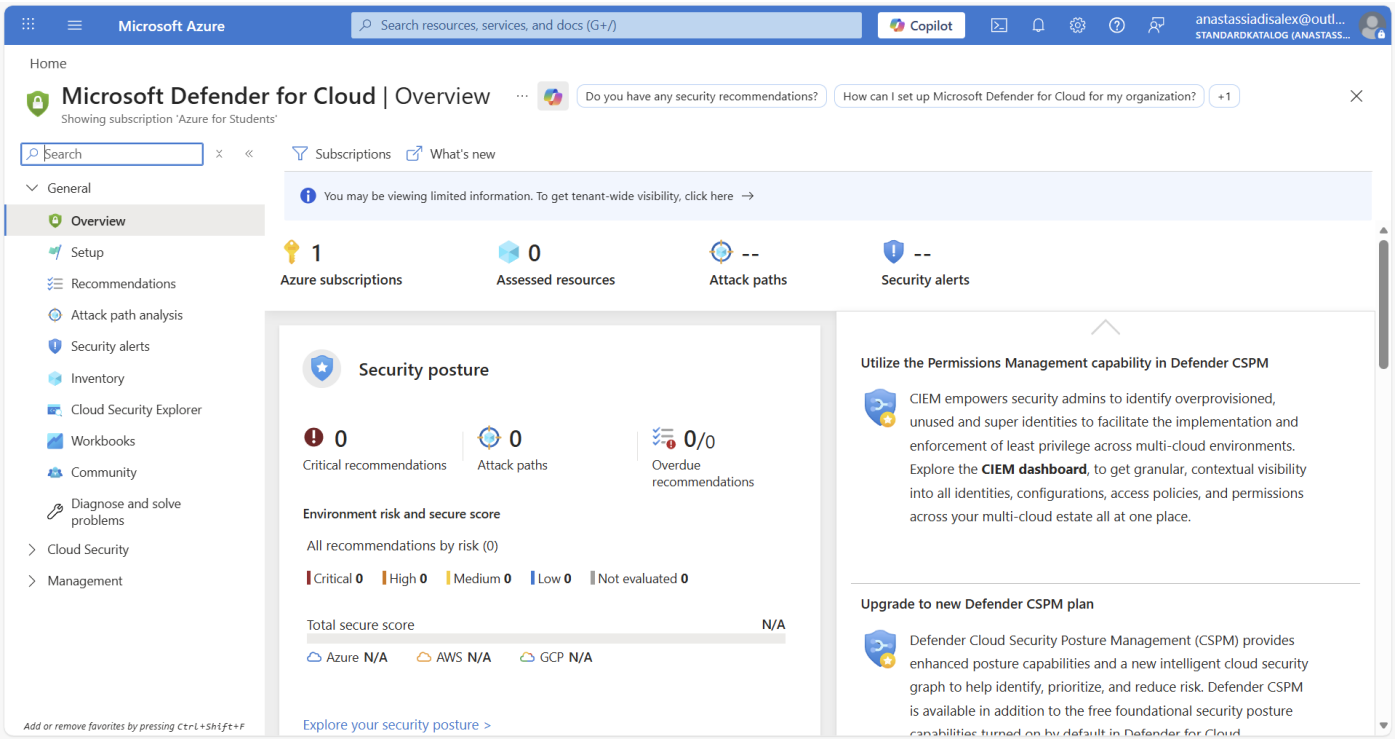

Also showing that the policy is compliant




This policy is demonstrating governance control.

*Step 2.3 – Defender for Cloud*

Image on defender for cloud overview (BEFORE)




Enabled (Defender for Servers) in Microsoft Defender for Cloud

Made a test VM (resources must exist to get recommendations)

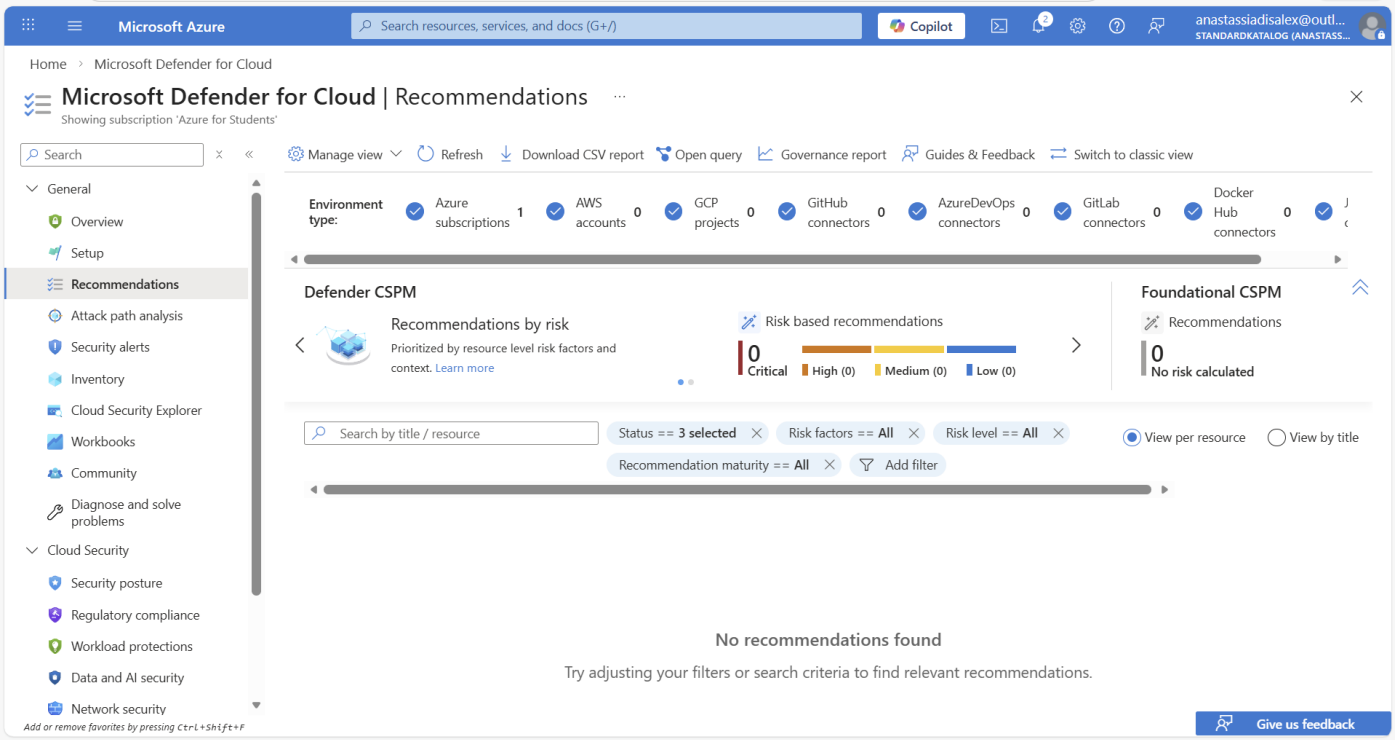
After provisioning and assessment:

- No critical or high-risk recommendations were identified
- No publicly exposed management ports detected
- VM baseline configuration aligned with security best practices

This confirms:

- Preventive controls (Azure Policy) are effective
- Network exposure is restricted
- Baseline security posture is hardened

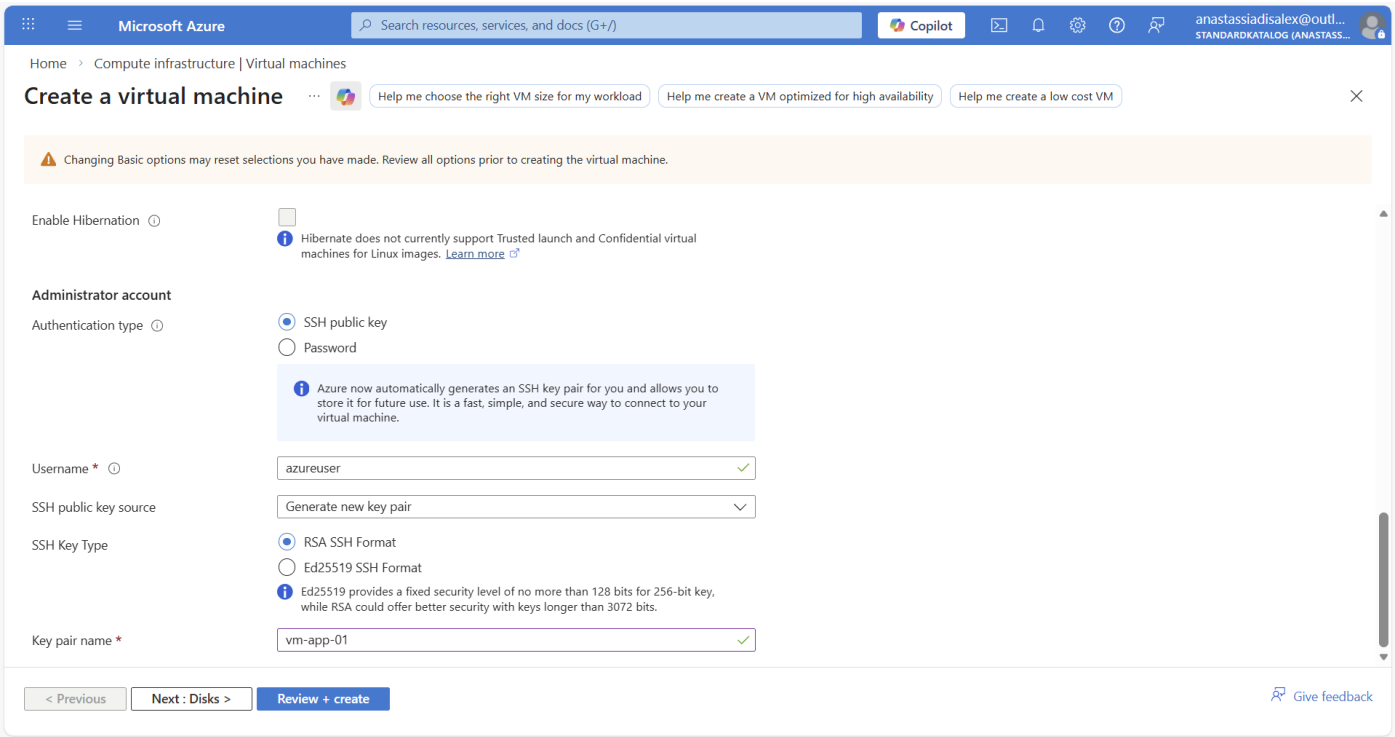
Secure Score remained stable with no high-risk findings.



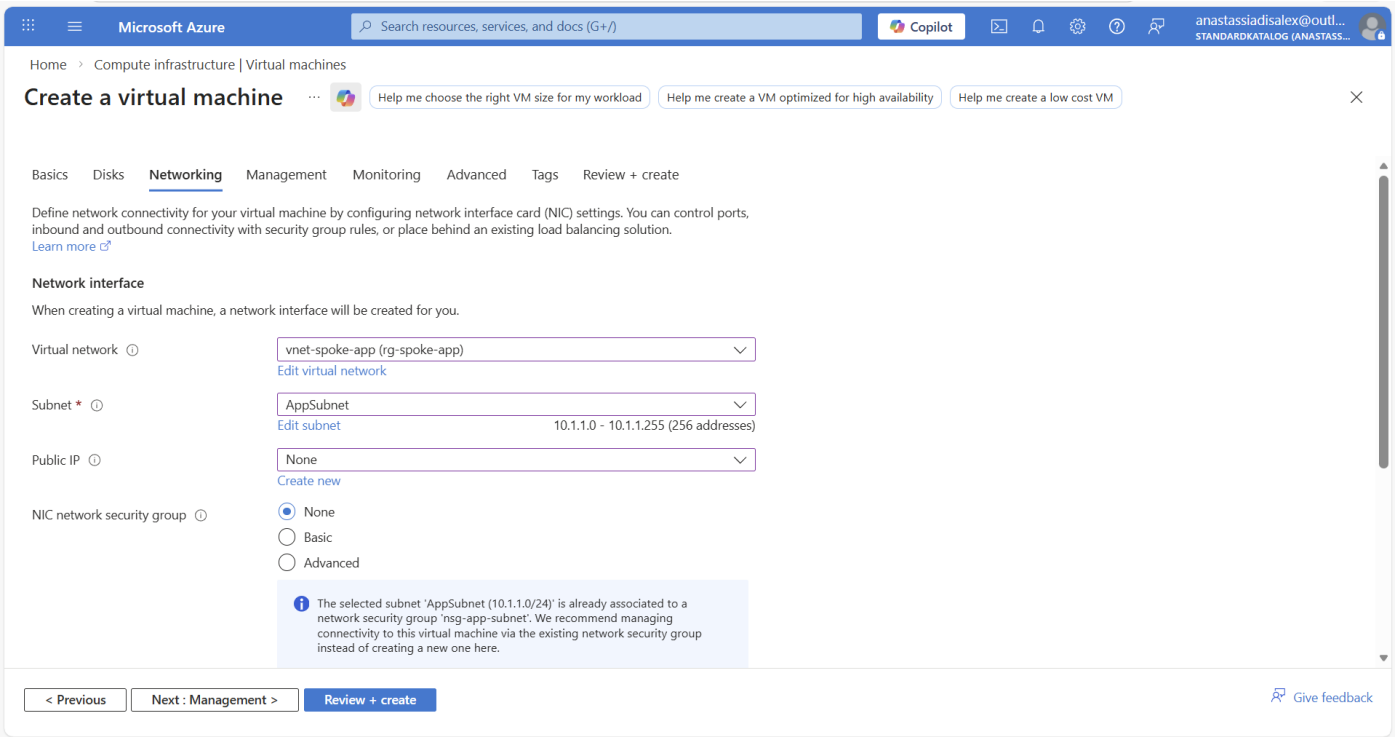

## Phase 3 – Secure VM

*Step 3.1 - Create a secure VM (Linux)*

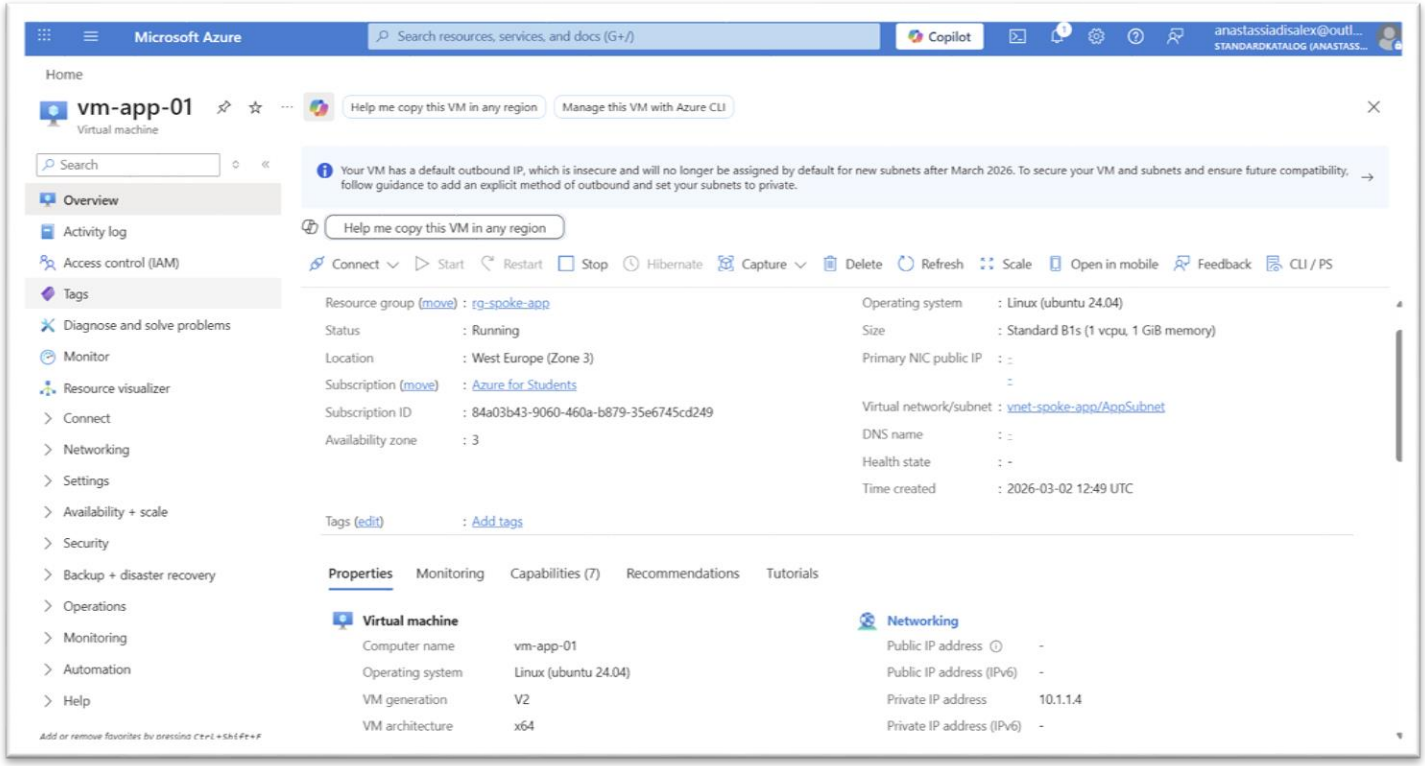
- No password authentication
- No public port exposure
- Secure-by-design deployment



- Public IP set to none
- Set to the right VNet as well as the right Subnet



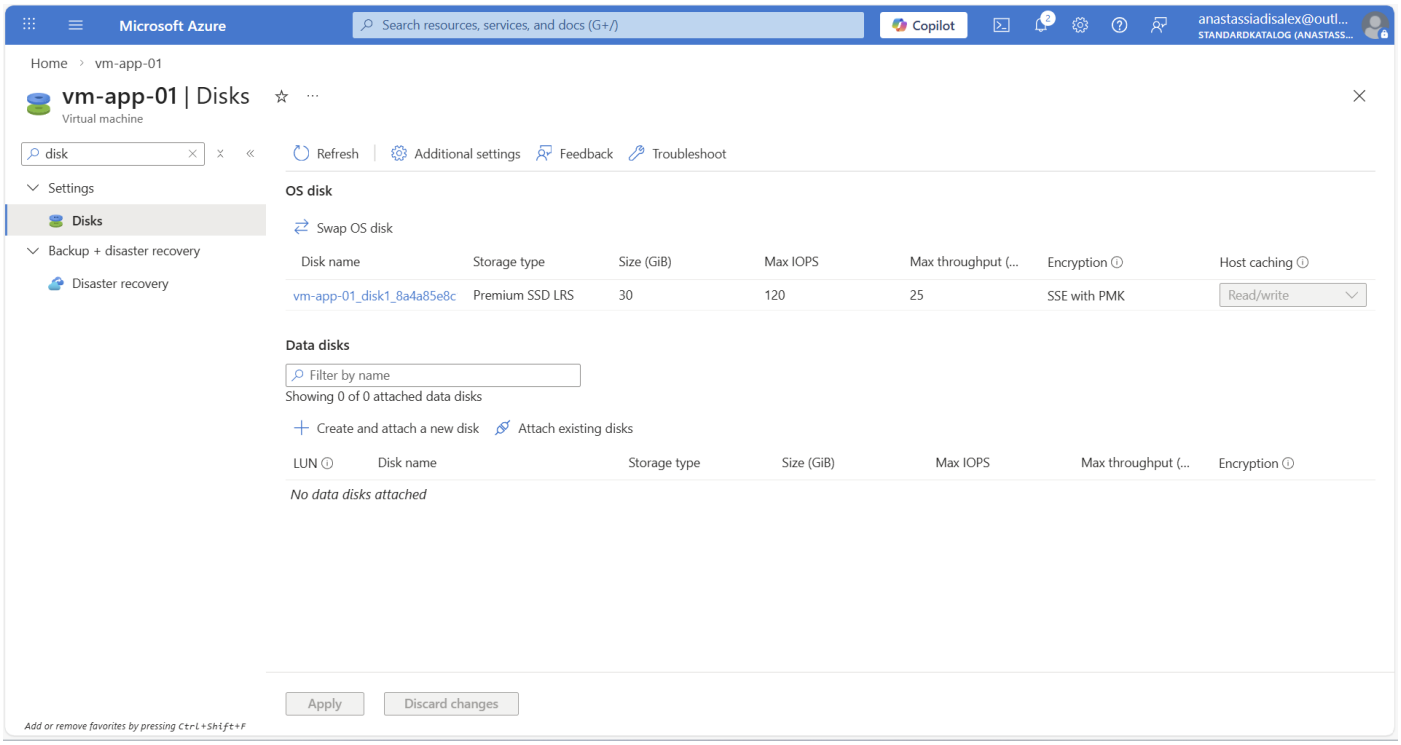

- VM overview page:




All these precautions are amplifying zero direct internet exposure.

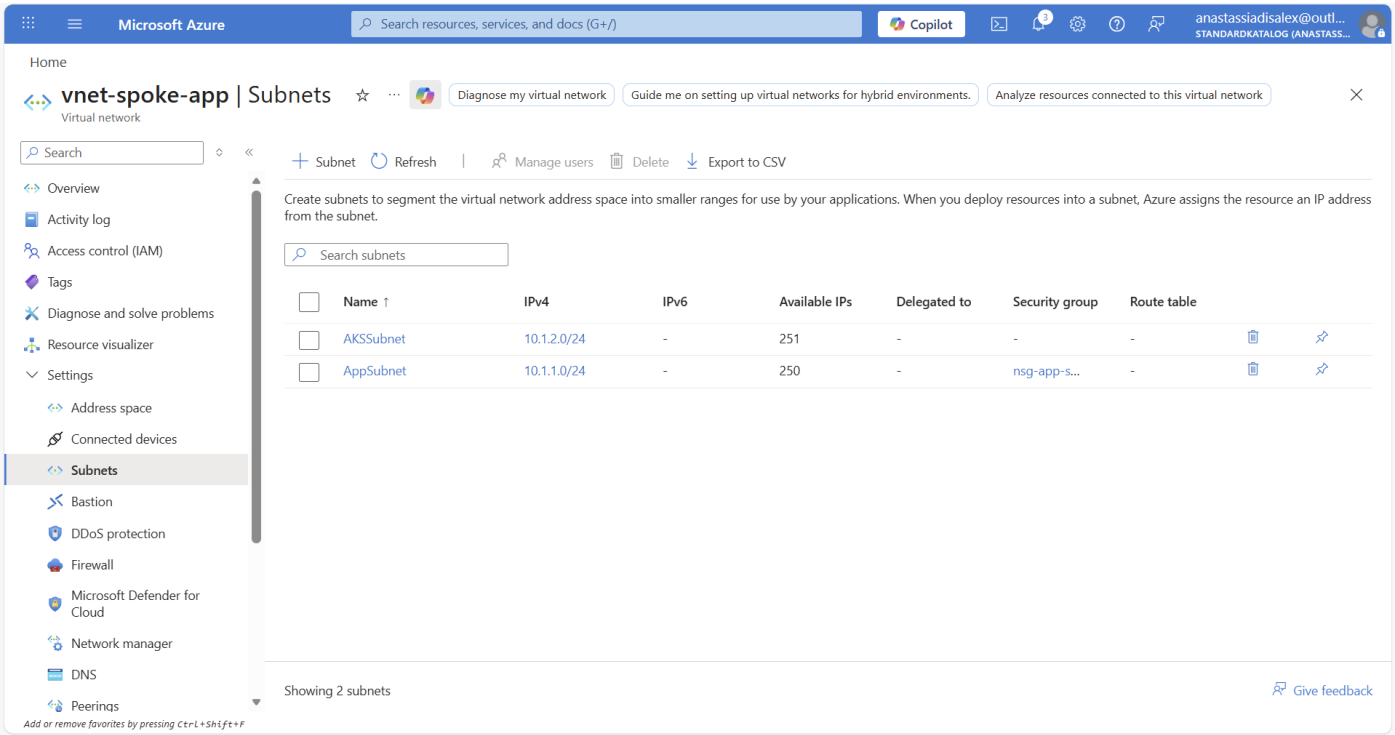
*Step 3.2 - Verify Disk Encryption*

- Proves data at rest protection
- Encryption is shown as Azure Storage Service Encryption (SSE) with Platform Managed Keys (PMK).




*Step 3.3 - Confirm Network Segmentation*

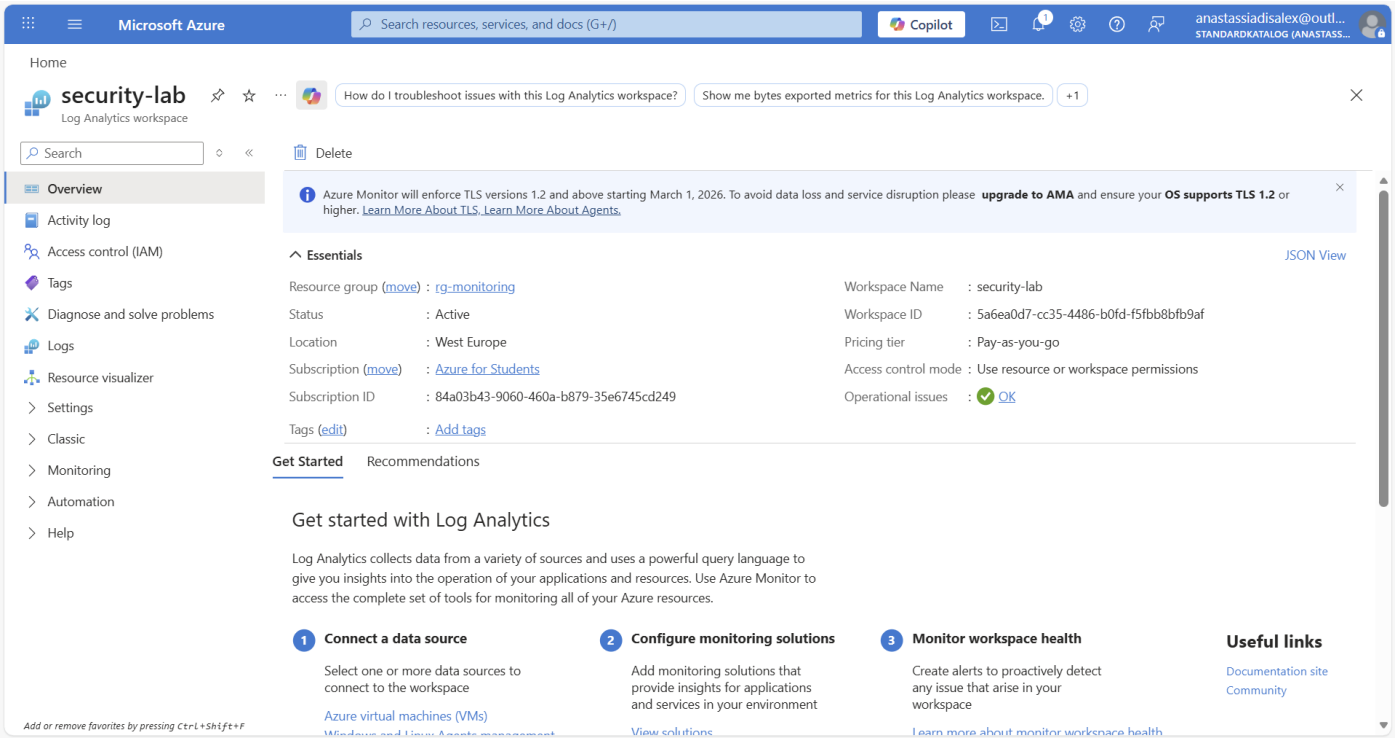
AppSubnet is attached to the NSG (nsg-app-subnet) enforcing subnet level network restrictions.




## Phase 4 – Monitoring & Logging

*Step 4.1 - Create Log Analytics Workspace*

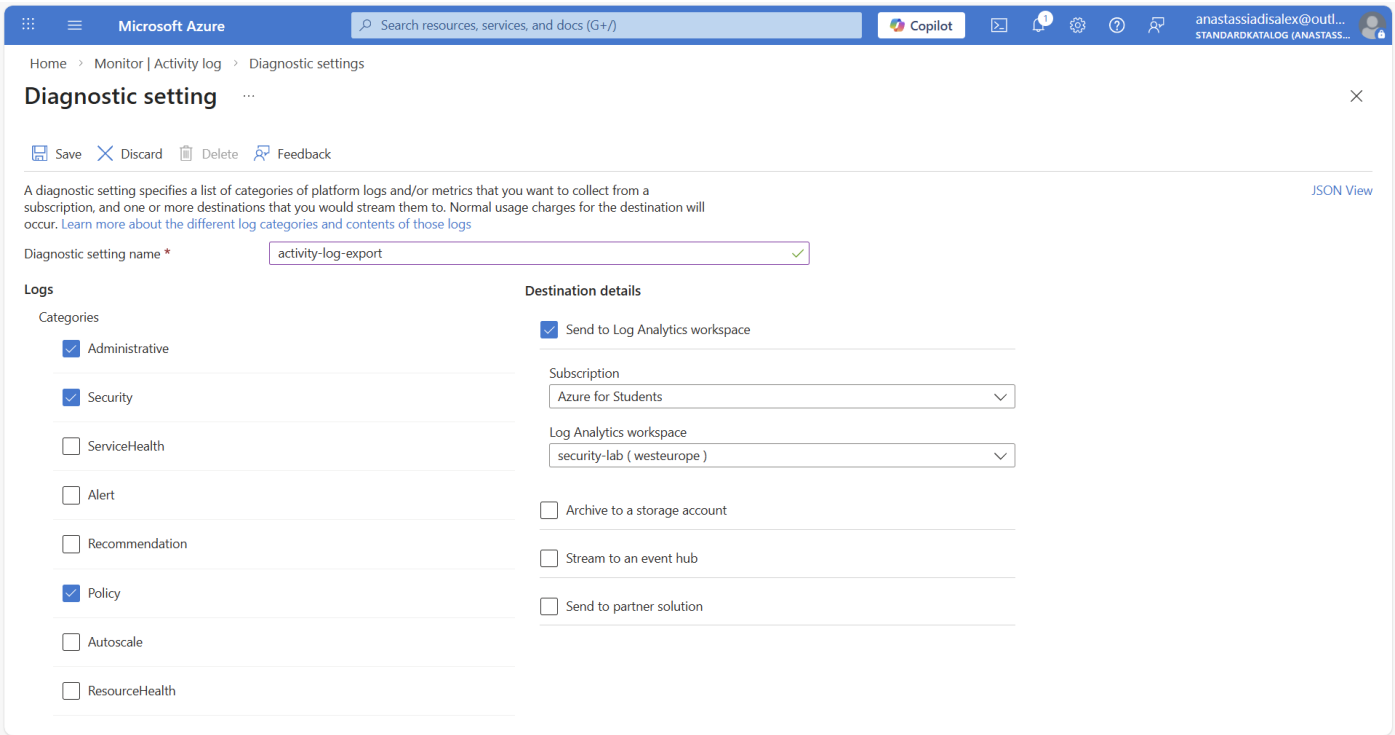
Log Analytic Workspace was created for monitoring and logging across the environment.




*Step 4.2 - Activate (Activity Log Export)*

Check the Administrative, Security & Policy box as well as Send to Log Analytics workspace with the right subscription, and the workspace created in the 4.1 step.

This is governance level tracking and auditing for different events.

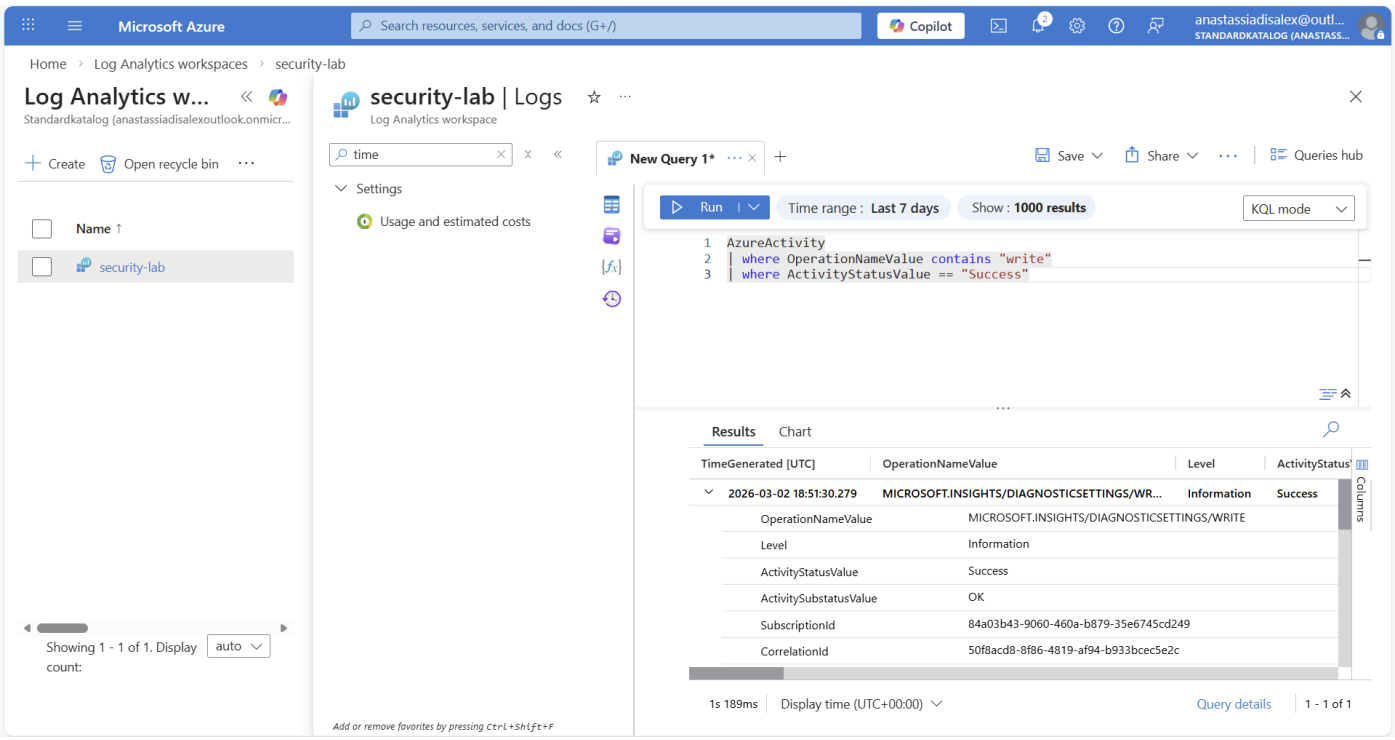



*Step 4.4 - Run KQL Query*

The runned Query:

```
AzureActivity
| where OperationNameValue contains "write"
| where ActivityStatusValue == "Success"
```

The query returned successfull write operations.




**OVERALL STRUCTURE SUMMARY:**

This environment was designed using a secure enterprise architecture model:

- Hub-and-Spoke networking
- Zero public exposure for workloads
- Enforced governance via Azure Policy
- RBAC with least privilege
- Defender for Cloud posture validation
- Centralized logging and monitoring

The lab demonstrates preventive, detective, and governance security controls aligned with Azure best practices.